

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Changes in Training FSM on Compliance Port Operations

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

In Gen 4 Electrical Compliance modes, the timing requirements in TS1 and TS2 sub-states are not relevant. The Indication field in TS3 should be 6h when in Compliance mode. Adding definitions of TX and RX Compliance modes. Removing Connection Manager note about exiting Compliance mode by putting the Router in uninitialized state.
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Benefits as a result of the changes:

Allows simpler Compliance flows.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). Table 4-33 – Transmitter Behavior in Gen 4 Training Sub-states To Text:

State	Transmitter Behavior
TS1	Transmitters shall send back-to-back Gen 4 TS1. The <i>Indication</i> field shall be 1h while any receiver is not ready for PAM3. <i>Indication</i> field shall be 2h when all enabled receivers are ready for PAM3.
TS2	Transmitters shall send back-to-back Gen 4 TS2. The <i>Indication</i> field shall be 3h while any receiver is executing the TxFFE negotiation. <i>Indication</i> field shall be 4h after all enabled receivers complete TxFFE negotiation.
TS3	Transmitters shall send back-to-back Gen 4 TS3. If not in Compliance mode, the <i>Indication</i> field shall be 5h before all enabled receivers detect Gen 4 TS3. The <i>Indication</i> field shall be 6h <u>when in Compliance mode or</u> after all enabled receivers detect Gen 4 TS3.
TS4	Transmitters shall send back-to-back Gen 4 TS4. The <i>Counter</i> field value shall be 0h before activating SSC and tSSCActivated time after activating SSC. The <i>Counter</i> field shall be incremented from 1h to Fh on each Gen 4 TS4, starting simultaneously on all enabled transmitters, tSSCActivated time after SSC is activated. Transmitters shall activate SSC within tActivateSSC time after sending the first trit of the first TS4.

(b). 4.2.1.3.2.2 – Transmitter Behavior in Gen 4 Training Sub-states To Text:

When the Port is not in RX Compliance mode or TX Compliance mode it shall meet the following timing requirement:

- ~~A USB4 Port shall t~~Transition to the TS2 sub-state within tGen4TS1 time after entering the TS1 sub-state.
- ~~A USB4 Port shall t~~Transition to the TS3 sub-state within tGen4TS2 time after entering the TS2 sub-state.

(c). 8.3.2.2.1 – SET_TX_COMPLIANCE (Required) To Text:

After receiving a SET_TX_COMPLIANCE Port Operation, a USB4 Port is in TX Compliance mode and shall disable the transition from Training state to CLd state due to timeouts defined in Section 4.2.1.3.3. When the USB4 Port detects SBRX at logical low for tDisconnectRx, it is no longer in TX Compliance mode, and it shall re-enable the transition from Training state to CLd state due to timeouts.

(d). 8.3.2.2.2 – SET_RX_COMPLIANCE (Required) To Text:

After receiving a SET_RX_COMPLIANCE Port Operation, a USB4 Port is in RX Compliance mode and shall disable the transition from Training state to CLd state due to timeouts defined in Section 4.2.1.3.3. When the USB4 Port detects SBRX at logical low for tDisconnectRx, it is no longer in RX Compliance mode, and it shall re-enable the transition from Training state to CLd state due to timeouts. After receiving a SET_RX_COMPLIANCE Port Operation, A USB4 Port operating at Gen 4 speed shall enable its receivers without requiring an LFPS handshake.

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(e). 8.3.2.2.4 – END_BER_TEST (Required)

To Text:



CONNECTION MANAGER NOTE

After a USB4 Port executes the END_BER_TEST Operation, the Connection Manager can bring the USB4 Port out of compliance mode by transitioning the Router with the USB4 Port to the Uninitialized state.

(f). 8.3.2.2.5 – END_BURST_TEST (Conditional)

To Text:



CONNECTION MANAGER NOTE

After a USB4 Port executes the END_BURST_TEST Operation, the Connection Manager can bring the USB4 Port out of compliance mode by transitioning the Router with the USB4 Port to the Uninitialized state.